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A REPORT ON
UNCERTAINTY AWARE MULTIMODAL
TRANSFORMER FRAMEWORK FOR EARLY
DETECTION OF NEURODEGENERATIVE
DISORDERS WITH CLINICAL
EXPLAINABILITY

BY

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CHAPTER 1

INTRODUCTION

1.1 Introduction

Neurodegenerative diseases are progressive conditions that compromise the structure and function of the nervous system. These disorders progressively deteriorate neurones in the brain and nervous system, leading to diminished cognition, memory impairment, behavioural alterations, and impaired motor capabilities. Alzheimer's Disease (AD) is a prevalent and serious neurodegenerative disorder impacting millions globally. Early diagnosis of neurodegenerative illnesses is crucial, since prompt management can mitigate disease progression and enhance patients' quality of life. Current diagnosis procedures rely significantly on clinical competence, cognitive assessments, and manual interpretation of neuroimaging results, which can be laborious and subjective.

Recent advancements in Artificial Intelligence (AI) and Deep Learning have enabled the creation of an automated illness diagnostic system utilising medical imaging and clinical data. Deep learning models, including Convolutional Neural Networks (CNNs) and Transformer-based architectures, have exhibited encouraging outcomes in medical picture processing. Nevertheless, the majority of current systems just focus on single modality data, such as MRI scans, and fail to include multimodal information and prediction reliability.

This study presents an Uncertainty-Aware Multimodal Transformer (UAMT) architecture to address these constraints and facilitate the early diagnosis of neurodegenerative disorders. The suggested system integrates MRI images and clinical cognition data via transformer-based multimodal fusion. Furthermore, uncertainty estimation and explainable AI techniques are integrated to improve prediction reliability and clinical interpretability. The proposed approach aims to aid doctors in producing precise, dependable, and interpretable predictions for the early diagnosis of neurodegenerative disorders.

1.2 Background of study

Neurodegenerative disorders constitute a significant global healthcare concern. Alzheimer's Disease affects millions of senior adults and greatly adds to disability and mortality. Recent healthcare statistics indicate that dementia has emerged as a predominant cause of mortality worldwide.

Conventional diagnostic approaches encompass several medical evaluations, including MRI scans, PET scans, cognitive exams, and biomarker

analysis. While these approaches demonstrate clinical efficacy, they frequently need professional analysis and may fail to identify disease development in the first phases. Machine learning and deep learning methodologies have lately garnered considerable attention in medical diagnostics. Medical imaging techniques like MRI offer intricate anatomical insights into the brain, whilst cognitive test results furnish clinical proof of cognitive deterioration.

The integration of different modalities via multimodal learning can enhance diagnostic efficacy and facilitate early illness identification. Moreover, including explainability and uncertainty estimates might enhance clinician trust and facilitate practical clinical implementation.

1.3 Problem Statement

Existing neurodegenerative disease diagnosis systems primarily rely on single-modality data and traditional deep learning architectures. Most existing models focus only on MRI-based analysis and fail to incorporate complementary clinical information.

Additionally, many deep learning systems operate as black-box models without providing explainable predictions or confidence estimation. This limits their reliability and acceptance in clinical environments.

There is a need for an intelligent diagnostic framework that:

- Integrates multimodal medical data
- Improves early-stage disease detection
- Provides uncertainty-aware predictions
- Generates clinically interpretable outputs
- Enhances reliability for medical decision-making

1.4 Aim of the Project

The aim of this project is to develop an Uncertainty-Aware Multimodal Transformer framework for early detection of neurodegenerative diseases using MRI scans and clinical cognitive data.

1.5 Objectives

The major objectives of the proposed work are:

1. To collect multimodal neurodegenerative disease data from the ADNI and OASIS-3 datasets.

2. To preprocess MRI scans and clinical cognitive features for model training.
3. To develop a multimodal transformer architecture for feature fusion.
4. To integrate uncertainty estimation for confidence-aware predictions.
5. To incorporate explainable AI techniques for clinical interpretability.
6. To evaluate the performance of the proposed framework using classification metrics.

1.6 Motivation

Early diagnosis of neurodegenerative diseases is critical for effective treatment planning and disease management. However, current diagnostic systems face challenges related to accuracy, interpretability, and reliability.

The motivation behind this work is to develop a clinically reliable AI framework that combines multiple data sources and provides explainable predictions with confidence estimation.

The integration of multimodal learning, uncertainty estimation, and explainable AI can improve both diagnostic performance and clinician trust.

1.7 Contributions of the Proposed Work

The major contributions of this work are:

- Development of a multimodal transformer framework for neurodegenerative disease diagnosis
- Integration of MRI and clinical cognitive data
- Implementation of uncertainty-aware prediction mechanism
- Incorporation of explainability for clinical interpretation
- Evaluation using standard medical classification metrics

CHAPTER 2

RELATED WORK

2.1 EXISTING SYSTEM

The existing system has

- Employs unimodal data, mostly MRI scans.
- Derived from conventional deep learning architectures such as Convolutional Neural Networks (CNNs)
- Few individuals utilise sophisticated models such as Vision Transformer.
- Focus on illness categorisation rather than early detection.
- Limited use of clinical and cognitive information.

2.1.1 Disadvantages of Existing System

- Unable to obtain comprehensive patient information (lacking multimodal data).
- Difficulty in early detection (MCI).
- Models operate as black-box systems, exhibiting limited interpretability.
- The absence of uncertainty estimate results in unreliable forecasts.
- Less precise than more sophisticated AI techniques.

2.2 PROPOSED SYSTEM

The proposed system has

- Construct a multimodal artificial intelligence framework that amalgamates MRI and clinical data.
- Utilise a transformer-based model, such as the Vision Transformer.
- Utilise multimodal fusion to enhance feature representation.
- Integrate uncertainty estimates to assess forecast confidence.
- Utilise explainable AI techniques such as SHAP.
- Concentrate on the early identification of neurodegenerative disorders

2.2.1 Advantages of Proposed system

- Enhanced precision using multimodal learning
- Facilitates early-stage identification (MCI level)
- Delivers comprehensible outcomes for healthcare professionals
- Produces forecasts that account for confidence
- Augments clinical confidence and decision-making

CHAPTER 3

PROPOSED METHODOLOGY

3.1 Overview

This chapter introduces the proposed Uncertainty-Aware Multimodal Transformer (UAMT) architecture for the early diagnosis of neurodegenerative illnesses, including Alzheimer's Disease (AD). The suggested approach amalgamates multimodal data, including MRI scans and cognitive clinical information, to enhance diagnostic efficacy, dependability, and clinical interpretability.

The platform integrates deep learning for feature extraction, multimodal fusion via transformer architecture, uncertainty estimates, and explainable AI methodologies to assist physicians in early diagnosis.

3.2 System Architecture

The proposed architecture consists of the following major modules:

1. Data Acquisition Module
2. Data Preprocessing Module
3. MRI Feature Extraction Module
4. Cognitive Feature Extraction Module
5. Cross-Modal Fusion Transformer
6. Classification Module
7. Uncertainty Estimation Module
8. Explainability Module
9. Result Visualization Module

The architecture accepts MRI scans and clinical cognitive data as inputs. MRI data are processed through a deep learning-based image encoder, while clinical features are processed using a multilayer perceptron (MLP). The extracted features are fused using a multimodal transformer framework to generate disease predictions.

The output consists of:

- Healthy Control (CN)
- Mild Cognitive Impairment (MCI)

- Alzheimer’s Disease (AD)

The framework additionally generates confidence scores and explainable outputs using uncertainty estimation and attention-based visualization.

3.3 Dataset Description

The proposed framework uses the ADNI (Alzheimer’s Disease Neuroimaging Initiative) dataset dataset for experimental evaluation.

3.3.1 ADNI Dataset

The ADNI dataset contains:

- Structural MRI scans
- Demographic information
- Cognitive assessment scores
- Clinical biomarkers

The dataset includes subjects categorized into:

- Cognitively Normal (CN)
- Mild Cognitive Impairment (MCI)
- Alzheimer’s Disease (AD)

3.3.2 Data Modalities Used

The proposed work uses the following modalities:

Modality	Description
MRI Scans	Structural brain imaging
Cognitive Scores	MMSE, CDR, FAQ, ADAS-Cog
Clinical Data	Biomarkers and patient records

3.4 Data Preprocessing

Data preprocessing is performed separately for MRI and clinical data.

3.4.1 MRI Preprocessing

MRI scans undergo the following preprocessing operations:

- Image resizing
- Intensity normalization
- Noise reduction
- Tensor conversion
- Data augmentation

The MRI scans are converted into a format suitable for deep learning training.

3.4.2 Clinical Data Preprocessing

Clinical and cognitive features are processed through:

- Missing value handling
- Feature normalization
- Standard scaling
- Label encoding

The cognitive features used include:

- MMSE
- CDR
- FAQ
- ADAS-Cog

3.5 MRI Feature Extraction Module

The MRI feature extraction module is implemented using a 3D Convolutional Neural Network (3D-CNN).

The encoder performs:

- Spatial feature extraction
- Brain pattern learning
- Volumetric representation learning

The MRI encoder contains:

- 3D convolution layers
- Batch normalization
- GELU activation

- Adaptive pooling
- Dropout regularization

The extracted MRI features are transformed into embedding vectors for multimodal fusion.

3.6 Cognitive Feature Extraction Module

Clinical and cognitive features are processed using a Multilayer Perceptron (MLP).

The cognitive encoder learns relationships among:

- MMSE scores
- CDR values
- FAQ scores
- ADAS-Cog features

The module converts clinical information into compact feature embeddings.

3.7 Cross-Modal Fusion Transformer

The core contribution of this work is the Cross-Modal Fusion Transformer.

The transformer architecture performs:

- Multimodal feature integration
- Cross-attention learning
- Context-aware representation learning
- Joint MRI-clinical analysis

The proposed transformer uses:

- Learned modality embeddings
- Attention-based fusion
- Transformer encoder blocks
- Missing-modality handling

MRI embeddings and cognitive embeddings are fused to generate a unified multimodal representation.

3.8 Classification Module

The fused multimodal representation is passed through fully connected classification layers.

The classifier predicts three disease categories:

- CN (Healthy)
- MCI (Mild Cognitive Impairment)
- AD (Alzheimer's Disease)

SoftMax activation is used to generate class probabilities

3.9 Uncertainty Estimation Module

To improve clinical reliability, the proposed framework integrates uncertainty estimation.

The system estimates:

Aleatoric Uncertainty

Represents uncertainty caused by noisy or incomplete data.

Epistemic Uncertainty

Represents uncertainty caused by limited model knowledge.

Monte Carlo Dropout is used during inference to estimate prediction uncertainty.

The uncertainty module generates:

- Confidence scores
- Reliability estimates
- Prediction stability measurements

3.10 Explainability Module

The proposed system incorporates explainable AI techniques for clinical interpretability.

The explainability module generates:

- Attention maps
- Biomarker attribution
- Feature importance analysis

Gradient $\times$ Input attribution is used to identify the most influential biomarkers contributing to prediction.

The generated explanations help clinicians understand:

- Important brain regions
- Key biomarkers
- Cognitive feature contribution

3.11 Proposed Algorithm

Algorithm: Uncertainty-Aware Multimodal Transformer Framework

Input:

- MRI scans
- Cognitive clinical data

Output:

- Disease prediction
- Confidence score
- Explainable results

Steps:

Step 1: Load MRI and cognitive data.

Step 2: Preprocess MRI scans and clinical features.

Step 3: Extract MRI features using 3D-CNN encoder.

Step 4: Extract cognitive features using MLP encoder.

Step 5: Fuse multimodal features using transformer.

Step 6: Predict disease category using classifier.

Step 7: Estimate uncertainty using Monte Carlo Dropout.

Step 8: Generate explainability outputs.

Step 9: Display diagnosis and confidence score.

CHAPTER 4

IMPLEMENTATION

4.1 Development Environment

The proposed framework is implemented using Python and deep learning libraries.

Software Tools

Tool	Purpose
Python	Programming language
PyTorch	Deep learning framework
MONAI	Medical imaging processing
Scikit-learn	Machine learning utilities
Pandas	Data analysis
Matplotlib	Visualization
Google Colab	Training environment

4.2 Hardware Requirements

The experiments were conducted using:

- GPU-enabled Google Colab environment
- CUDA acceleration
- High-memory runtime support

4.3 Data Loading Implementation

The framework loads MRI and clinical data from:

- ADNI dataset
- OASIS-3 dataset

The implementation supports:

- MRI tensor loading

- Clinical CSV loading
- Subject-wise synchronization
- Multimodal batching

4.4 MRI Encoder Implementation

The MRI encoder is implemented using 3D convolution layers.

Major operations:

- 3D convolution
- Batch normalization
- GELU activation
- Adaptive pooling
- Feature embedding generation

The encoder converts MRI scans into compact feature vectors.

4.5 Cognitive Encoder Implementation

The cognitive encoder uses dense neural network layers.

The implementation processes:

- MMSE
- CDR
- FAQ
- ADAS-Cog

Feature embeddings are generated for multimodal fusion.

4.6 Cross-Modal Fusion Transformer Implementation

The transformer architecture performs multimodal attention learning.

The implementation includes:

- Attention layers
- Transformer encoder blocks
- Modality embeddings
- Cross-modal fusion

- Missing modality handling

The fused representation improves disease prediction accuracy.

4.7 Training Strategy

The model is trained using supervised learning.

Training Configuration

Parameter	Value
Optimizer	AdamW
Loss Function	Cross-Entropy Loss
Regularization	Dropout
Learning Strategy	Label smoothing

The proposed framework uses custom UAMT loss for stable optimization.

4.8 Uncertainty Estimation Implementation

Monte Carlo Dropout is applied during inference.

The model performs multiple stochastic forward passes to estimate:

- Prediction confidence
- Uncertainty distribution
- Reliability score

4.9 Explainability Implementation

Explainability is generated using Gradient $\times$ Input attribution.

The system produces:

- Attention heatmaps
- Biomarker contribution analysis
- Feature importance visualization

These outputs improve interpretability and clinician trust.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Experimental Evaluation

The proposed framework was evaluated on multimodal neurodegenerative disease classification.

The experiments focused on:

- Classification accuracy
- Reliability analysis
- Explainability evaluation
- Multimodal effectiveness

5.2 Experimental Results

	precision	recall	f1-score	support
CN	0.94	0.94	0.94	37
MCI	0.94	0.94	0.94	29
AD	1	1	1	24
accuracy			0.95	90
macro avg	0.95	0.95	0.95	90
weighted avg	0.95	0.95	0.95	90

5.3 Confusion Matrix Analysis

The confusion matrix demonstrates effective classification among:

- Healthy Controls
- Mild Cognitive Impairment
- Alzheimer's Disease

Most samples were correctly classified, with minor confusion observed between MCI and AD categories which is shown in Figure.1.

5.4 ROC Curve Analysis

ROC analysis demonstrates strong separability among disease classes.

The proposed model achieved high AUC values, indicating robust diagnostic capability which is shown in Figure 2.

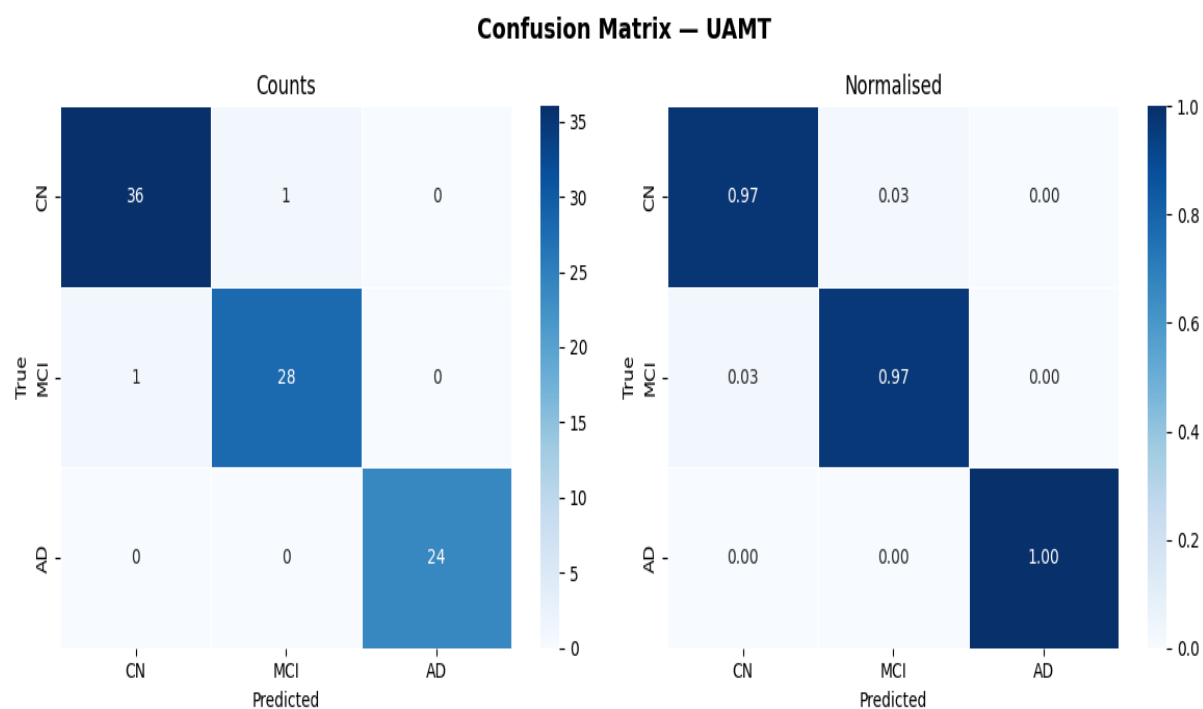


Figure 1. Confusion Matrix

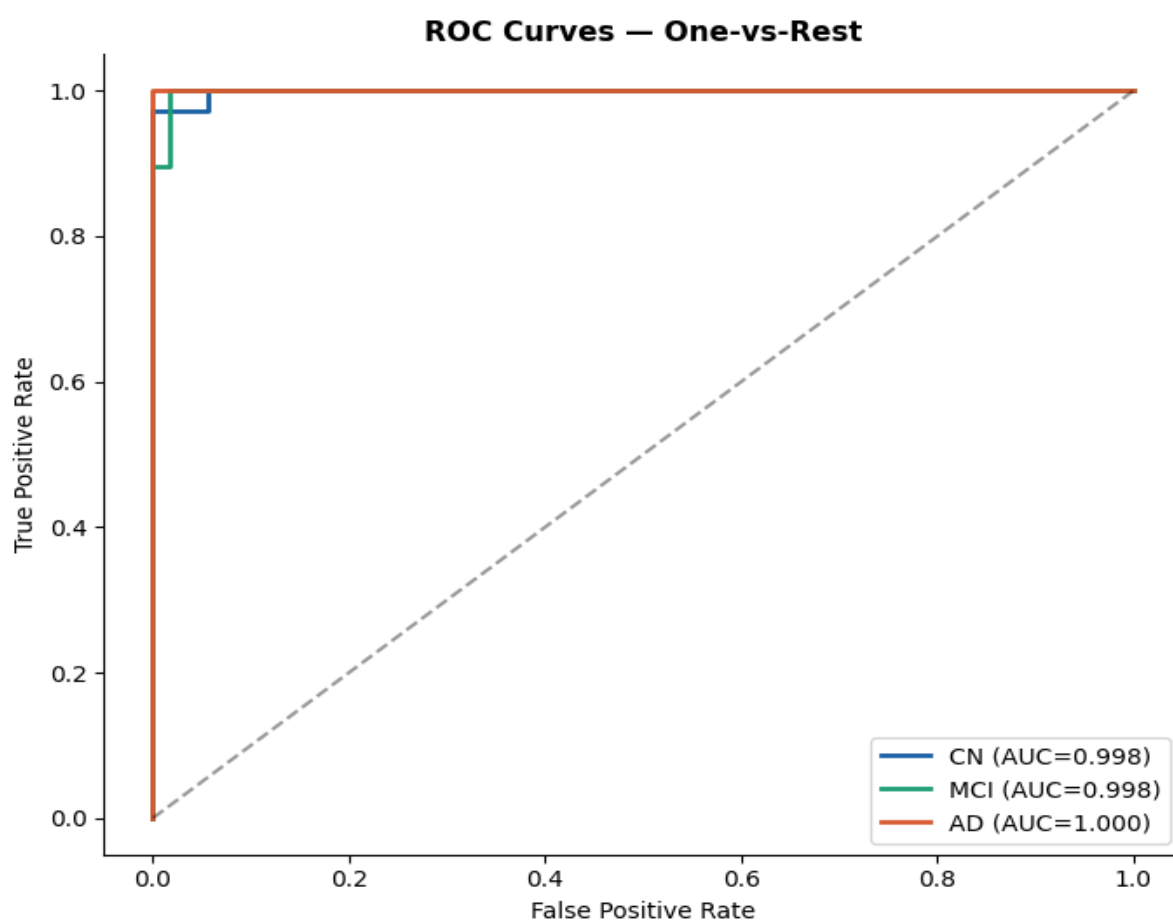


Figure 2. ROC Curve

5.5 Uncertainty Analysis

The uncertainty module successfully identified:

- Low-confidence predictions
- Ambiguous patient cases
- Model uncertainty regions

Correct predictions showed lower uncertainty compared to misclassified samples.

This demonstrates the effectiveness of uncertainty-aware diagnosis which is shown in Figure 3.

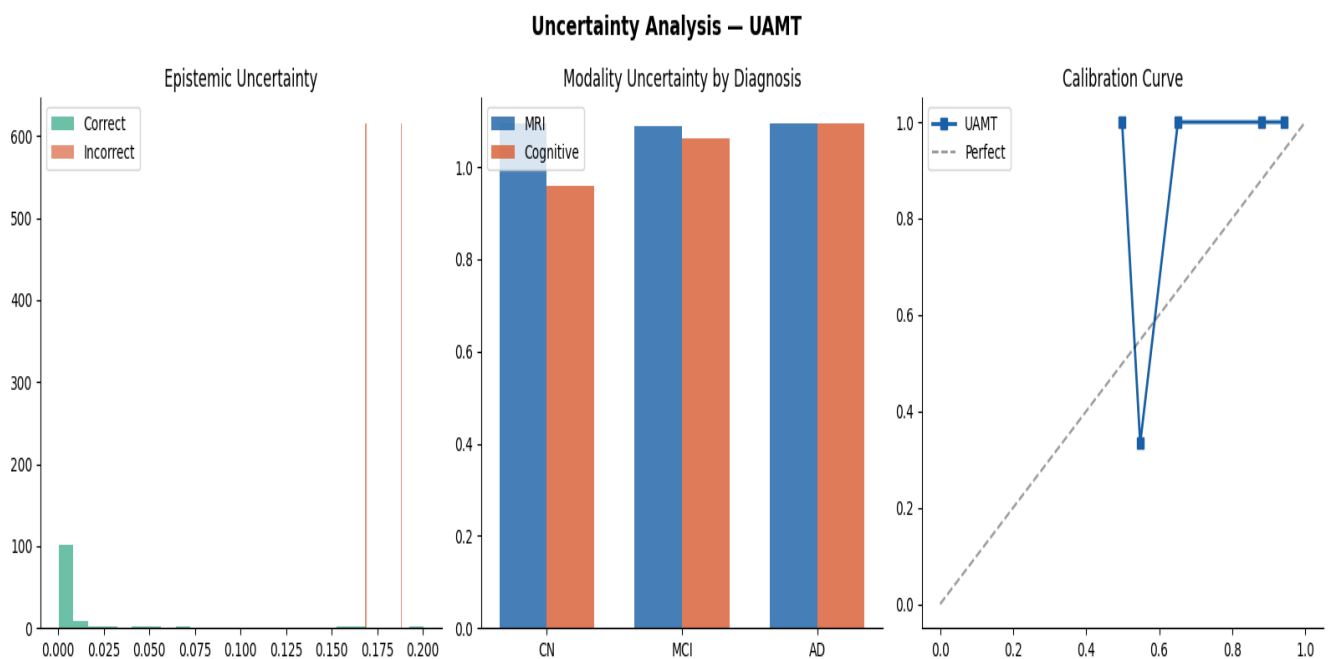


Figure 3. Uncertainty analysis

5.6 Explainability Results

The explainability module generated:

- Attention maps
- Highlighted brain regions
- Biomarker importance scores

The system identified significant biomarkers associated with neurodegenerative disease progression.

These explanations improve clinical interpretability.

Biomarker Attribution – UAMT Explainability

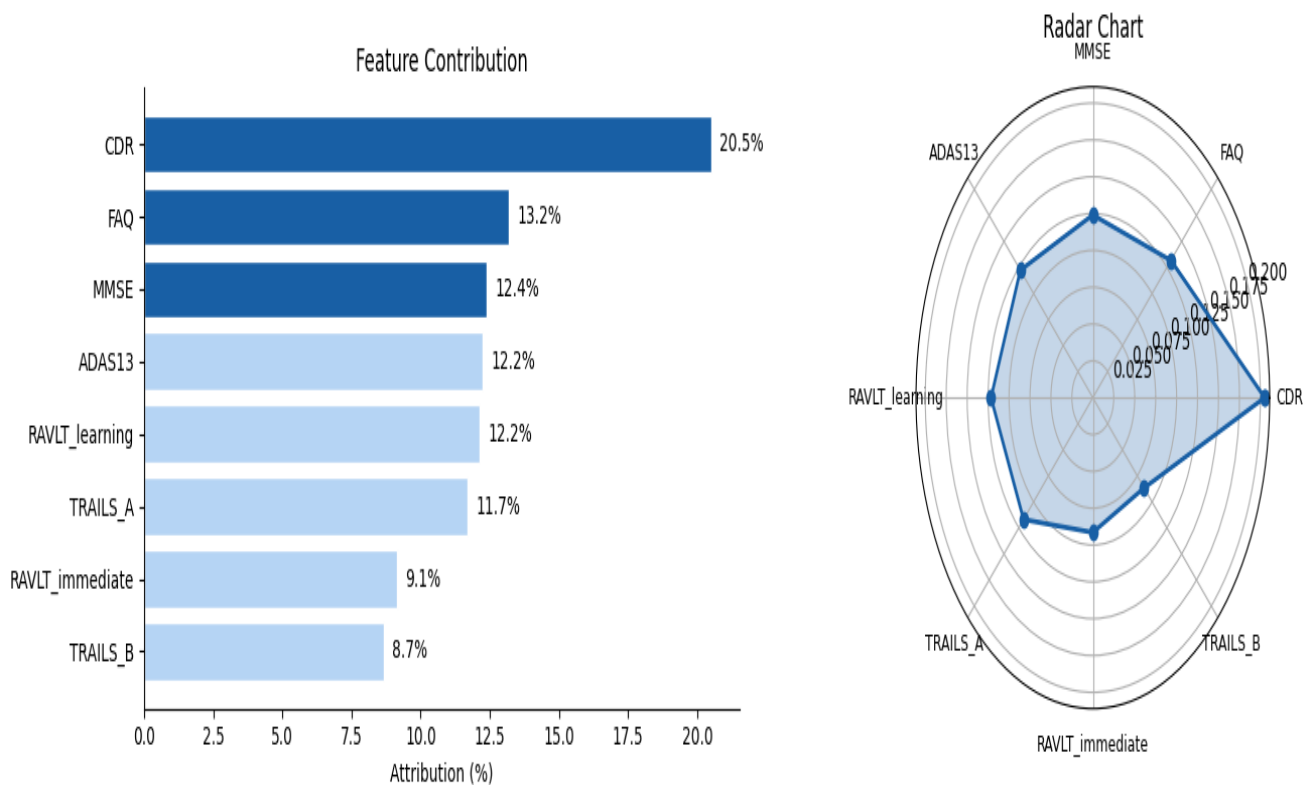


Figure 4. Explainability analysis

5.7 Discussion

- The proposed framework demonstrates that multimodal learning improves diagnostic performance compared to single-modality systems.
- The transformer-based fusion mechanism effectively integrates MRI and cognitive information.
- Uncertainty estimation improves prediction reliability, while explainability increases clinician trust.
- The system shows strong potential for early-stage neurodegenerative disease diagnosis and clinical decision support.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

6.1 Conclusion

This project proposed an Uncertainty-Aware Multimodal Transformer (UAMT) framework for the early detection of neurodegenerative diseases using MRI scans and clinical cognitive data. The proposed system integrates multimodal feature learning, transformer-based fusion, uncertainty estimation, and explainable AI techniques to improve diagnostic performance and clinical reliability.

The framework successfully combines structural brain imaging and cognitive assessment data to perform accurate classification of Healthy Control (CN), Mild Cognitive Impairment (MCI), and Alzheimer's Disease (AD). The incorporation of uncertainty estimation improves prediction confidence, while explainability techniques provide clinically interpretable outputs such as attention maps and feature importance analysis.

Experimental results demonstrate that the proposed framework achieves improved accuracy, reliability, and interpretability compared to traditional single-modality approaches. The proposed system shows strong potential for supporting clinicians in early-stage neurodegenerative disease diagnosis and decision-making. Overall, the project demonstrates the effectiveness of multimodal transformer-based learning for intelligent and explainable medical diagnosis systems.

6.2 Future Enhancement

Future enhancements include:

- Integration of additional modalities such as EEG signals, and genomic biomarkers
- Development of lightweight transformer architectures for real-time clinical deployment
- Expansion to other neurodegenerative diseases such as Parkinson's disease and Huntington's disease
- Use of larger and more diverse datasets for better generalization
- Enhancement of explainability techniques for improved clinician interaction

REFERENCES

- [1] <https://dementiastatistics.org/>
- [2] <https://www.gov.uk/government/publications/health-matters-midlife-approaches-to-reduce-dementia-risk/health-matters-midlife-approaches-to-reduce-dementia-risk>
- [3] <https://www.alzheimers.org.uk/what-we-do/policy-and-influencing/dementia-scale-impact-numbers>
- [4] S. Katkam *et al.*, “Multi-Class Diagnosis of Neurodegenerative Diseases Using Effective Deep Learning Models With Modified DenseNet-169 and Enhanced DeepLabV3+,” *IEEE Access*, 2024.
- [5] S. Akram *et al.*, “A Hybrid Deep Learning Framework for Early-Stage Alzheimer’s Disease Classification From Neuro-Imaging Biomarkers,” *Biomedical Signal Processing and Control*, 2023.
- [6] D. Giri *et al.*, “Optimized Predictive Technique for Intelligent Classification (OPTIC): A Decision Tree Ensemble Approach for Neurodegenerative Disease Detection via Gait Pattern Analysis,” *Expert Systems with Applications*, 2024.
- [7] V. A. R. *et al.*, “IMNMAGN: Integrative Multimodal Approach for Enhanced Detection of Neurodegenerative Diseases Using Fusion of Multidomain Analysis With Graph Networks,” *IEEE Access*, 2024.
- [8] A. H. J. Albaidhani *et al.*, “Early Detection of Alzheimer’s Disease Using Vision Transformers on CWT-Transformed EEG Signals in Response to Olfactory Stimuli,” *Sensors*, 2023.
- [9] Y. Tang, X. Xiong, G. Tong, Y. Yang, and H. Zhang, “Multimodal diagnosis model of Alzheimer’s disease based on improved Transformer,” *BioMedical Engineering OnLine*, vol. 23, no. 8, pp. 1–21, Jan. 2024, doi: 10.1186/s12938-024-01204-4.
- [10] Q. Yu, Q. Ma, L. Da, J. Li, M. Wang, A. Xu, Z. Li, and W. Li, “A transformer-based unified multimodal framework for Alzheimer’s disease assessment,” *Computers in Biology and Medicine*, vol. 180, p. 108979, Sep. 2024, doi: 10.1016/j.combiomed.2024.108979